

VIETNAM NATIONAL UNIVERSITY OF HO CHI MINH CITY
THE INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



Autonomous AI-Based Generation of Interactive Questions for MOOC Videos

By

NGUYEN TOAN PHUC

Student ID: ITITIU21093

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Autonomous AI-Based Generation of Interactive Questions for MOOC Videos

APPROVED BY:

Le Thanh Son, M.Sc., Supervisor

Tran Manh Ha, Assoc. Prof., Ph.D.

THESIS COMMITTEE

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ABSTRACT

The rapid growth of digital education has created a need for simple tools that help instructors build interactive learning content without specialist technical skills. At Vietnam National University – Ho Chi Minh City (VNU-HCM), creating H5P interactive videos through the WordPress ecosystem remains cumbersome and slow for many educators.

This thesis presents the design and evaluation of a dedicated platform for AI-assisted H5P video authoring. The system combines a React 18 and TypeScript frontend, a Node.js/Express backend, a SQLite database, and a transcript-driven AI pipeline that generates candidate H5P interactions for instructor review. The pipeline accepts caption sources from uploaded files, YouTube captions, or audio transcription, then validates each suggestion before it reaches the editor.

A comparative usability study with ten Information Technology students showed that the platform substantially improved authoring efficiency and usability over the WordPress baseline. Task completion increased from 30% to 95%, median creation time dropped from 45 minutes to 3.2 minutes, and usability scores rose from 2.1 to 4.8 on a five-point scale. The results indicate that combining user-centred design with generative AI can lower the barrier to creating interactive pedagogical material for university teaching.

Keywords: H5P, interactive video, educational technology, AI-assisted content authoring, usability evaluation, user-centred design.

CHAPTER 1

INTRODUCTION

1.1 Background

Over the past decade, Vietnamese higher education has undergone a profound structural shift driven by national digital transformation policy. Vietnam’s National Education Development Strategy 2021–2030, enacted under Decision No. 1373/QĐ-TTg, establishes concrete targets for the integration of digital technology into teaching and learning at all levels of the education system [3]. Institutions affiliated with Vietnam National University – Ho Chi Minh City (VNU-HCM) are at the forefront of this effort. The VNU-HCM MOOC platform, publicly available at <https://courses.vnuhcm.edu.vn>, has delivered thousands of course modules to learners across the country and serves as the primary digital learning infrastructure for the university [4].

Despite this progress, the gap between infrastructure availability and effective pedagogical use remains wide. Surveys conducted at HCMIU indicate that while 89% of instructors express interest in creating interactive digital content, only 23% report confidence in using current authoring tools. Faculty members cite interface complexity and the steep learning curve associated with plugin-based workflows as the primary barriers to adoption. This gap is not unique to Vietnam; comparative research on e-learning quality in developing university contexts consistently identifies technical usability as a decisive factor in adoption rates [5]. The challenge is therefore not a lack of will or institutional infrastructure, but a lack of purpose-built tools that meet educators where they are.

Interactive video has emerged as one of the most effective media for asynchronous and hybrid instruction because it combines the explanatory power of video with the engagement mechanisms of formative assessment [6]. Rather than passively consuming a recorded lecture, students interacting with embedded questions must retrieve and apply knowledge in real time, a mechanism that is well-supported by the evidence base for active learning [7]. Merkt et al. demonstrated that videos with interactive control features—particularly stop-and-answer moments—produced significantly better knowl-

edge retention and transfer scores than equivalent linear content when learners were actively engaged rather than passively watching [8]. Vural similarly found that embedding questions at strategic timestamps within educational videos improved post-test scores and reduced learner dropout in online environments [9].

The pedagogical design of these embedded questions is best guided by Bloom’s revised taxonomy, which provides a principled framework for targeting specific cognitive levels from basic recall (*remember*) through synthesis and evaluation (*create*) [10]. When questions are distributed across these levels and positioned at moments where a key concept has just been introduced, the extraneous cognitive load imposed by the interruption is minimised and germane cognitive load—the load associated with schema construction—is maximised [11–13].

H5P (HTML5 Package) is an open-source framework for creating, sharing, and embedding richly interactive content in web environments [14, 15]. Originally developed by Joubel AS in 2013, the H5P ecosystem has expanded to encompass over 50 content types, ranging from foundational assessments like Multiple Choice and True/False questions to sophisticated branching scenarios, and interactive timelines. A core design principle of the framework is that all content is self-contained: each H5P package exists as a single .h5p ZIP archive containing its own JavaScript libraries, media assets, and content parameters. This inherent portability allows H5P packages to be trivially imported into any H5P-aware Learning Management System (LMS), such as Moodle [16], Canvas, and Blackboard, via the LTI 1.3 standard [17].

H5P Interactive Video stands as the most widely utilized content type within higher education. The framework wraps a video source—whether a direct upload or a YouTube URL—in an H5P player that pauses playback at pre-defined timestamps to present interactive elements, requiring learner input before resuming. The specific interaction types supported and relevant to this implementation are Multiple Choice, True/False, Fill in the Blanks, Drag Text, Mark the Words, and Matching. Each of these types maps to a dedicated H5P library with a strictly defined content parameter schema, facilitating the programmatic generation of H5P content from structured JSON representations.

Despite H5P’s capabilities, the dominant deployment model in Vietnamese universities relies on the WordPress H5P plugin, which was designed for small-scale personal websites rather than institutional content production systems. Creating a single interactive video through this pathway requires an instructor to: (1) log in to a WordPress administration panel; (2) create or locate a “Post”; (3) activate the H5P shortcode block; (4) navigate to a separate H5P content editor; (5) upload or link the video; (6) locate each desired timestamp through a built-in player; (7) configure an interaction type with several nested settings screens; and (8) save and return to the post. This workflow spans multiple interfaces, involves concepts (posts, shortcodes, plugin settings) that lie entirely outside an educator’s pedagogical domain, and takes an average of 45 minutes for a first-time user

to complete even a simple three-question interactive video.

Research into educational technology adoption identifies the Technology Acceptance Model as a useful lens for understanding why educators reject tools [18]. Perceived ease of use and perceived usefulness together explain a substantial fraction of variance in adoption intention. When a tool imposes high extraneous cognitive load on every interaction—as the WordPress H5P workflow demonstrably does—perceived ease of use collapses, and adoption does not occur regardless of the tool’s underlying educational potential [11, 12].

The recent maturation of large language models (LLMs) has opened a new possibility for educational content authoring: the ability to analyse a video’s transcript and generate contextually appropriate, pedagogically structured interactive questions automatically [19, 20]. If an instructor can review and accept AI-generated suggestions rather than composing each question from scratch, the cognitive overhead of interactive video creation falls dramatically. The same pipeline that extracts a transcript, classifies its segments by topic and complexity, and returns a structured list of H5P interactions can also apply Bloom’s taxonomy labels to each suggested question, helping instructors achieve a deliberate spread of cognitive difficulty without specialist training in instructional design.

This thesis explores whether combining a purpose-built, user-centred authoring interface with an AI pipeline based on Anthropic Claude and a locally hosted Ollama inference server can produce a measurable improvement in educator productivity and satisfaction when creating H5P interactive videos.

1.2 Problem Statement

The central problem this thesis addresses is the mismatch between the educational potential of H5P interactive video and the practical usability of existing authoring environments for university instructors.

The problem manifests through four interacting challenges:

1. **Workflow complexity.** Current WordPress-based H5P workflows require educators to navigate at least six distinct screens and understand web publishing concepts that are irrelevant to their pedagogical task. Task analysis of the existing workflow measured 47 interface interactions across six screens to produce one interactive video, with 3.2-second median page-load times that interrupt creative flow.
2. **High cognitive load.** The extraneous cognitive load imposed by the technical interface reduces the mental resources available for pedagogical design [12, 21]. Instructors report that managing the tool consumes more attention than designing the questions themselves.
3. **Low adoption.** As a direct consequence of the above, adoption of H5P interactive

video at VNU-HCM has been limited to technically proficient faculty. Senior instructors and those in humanities disciplines are systematically excluded from using a tool that research demonstrates to be educationally effective [5, 18].

4. **Absence of AI assistance.** No existing freely available H5P authoring tool provides integrated AI assistance for question generation. Instructors who could benefit most from AI support—those with limited instructional design experience—have no mechanism for obtaining context-aware question suggestions tied to the content of their specific video.

The **primary research question** is therefore:

How can a purpose-built web platform combining user-centred interface design with an AI-driven transcript analysis pipeline improve the efficiency, quality, and adoption of H5P interactive video authoring by university instructors?

Four supporting research questions guide the investigation:

- RQ1. What are the specific usability barriers in current H5P authoring workflows that most strongly reduce task completion and satisfaction?
- RQ2. How can user-centred design principles be applied to reduce extraneous cognitive load in timestamp-based interactive question authoring?
- RQ3. What technical architecture best supports real-time AI-assisted question generation within an H5P authoring workflow?
- RQ4. To what extent does the purpose-built platform outperform the WordPress-based baseline on measurable usability and productivity metrics?

1.3 Scope and Objectives

The thesis pursues five concrete objectives in order to answer the research questions above.

1. **Platform development.** Design and implement a full-stack web application that enables instructors to upload or link a video, add H5P interactive elements at precise timestamps through a visual timeline interface, preview the resulting content in real time, and export a standards-compliant `.h5p` package suitable for direct import into the VNU-HCM MOOC infrastructure.
2. **AI pipeline integration.** Integrate a dual-provider AI pipeline that extracts a timestamped transcript from the uploaded video (via YouTube automatic captions, uploaded subtitle files, or Whisper audio transcription), sends the transcript to a large language model (Anthropic Claude or a locally hosted Ollama instance), and

returns a validated list of H5P interaction suggestions with timestamps, content type, question text, and a Bloom’s taxonomy classification.

3. **User-centred design.** Apply user-centred design methodology throughout the interface design process, grounding all design decisions in identified instructor needs and validated against usability heuristics.
4. **Security implementation.** Implement a security baseline aligned to the OWASP Top Ten, covering authentication, authorisation, input validation, transport security, and rate limiting.
5. **Empirical evaluation.** Conduct a comparative usability study measuring task completion rate, time-on-task, error rate, and System Usability Scale (SUS) score against the WordPress H5P baseline, and report the findings with statistical context.

The project scope is bounded as follows.

Content type scope. The platform supports the six H5P interaction types most relevant to interactive video pedagogy: Multiple Choice, True/False, Fill in the Blanks, Drag Text, Mark the Words, and Matching. All six are supported for manual authoring; the AI pipeline generates Multiple Choice, True/False, Fill in the Blanks, and Matching suggestions automatically, as these three types have deterministic schema structures that can be reliably validated via Zod.

Video source scope. The platform accepts direct video uploads in MP4, WebM, and MOV formats up to 500 MB, and also accepts YouTube video URLs. For YouTube-sourced videos, transcripts are extracted directly from the platform’s automatic captions. For uploaded video files, instructors may upload a WebVTT caption file; alternatively, the system can invoke **faster-whisper** locally to perform audio-to-text transcription.

User scope. The primary target user is a university instructor at VNU-HCM. System administration is handled by a second user role (Admin), which can manage user accounts, view usage analytics, and export audit logs.

Integration scope. The platform exports standard H5P packages that can be imported into any LTI-compatible LMS. An LTI 1.3 route is implemented in the backend for direct LMS launch and deep-linking. Student- level grade passback and learning-analytics reporting are not yet implemented and are treated as future work.

- **H5P content types.** The platform does not support the full catalogue of 50+ H5P content types. Branching scenarios, Course Presentation, and other complex types are out of scope.
- **Evaluation sample.** Usability testing was conducted with ten Information Technology students serving as educator proxies, not with actual university faculty. While this is a standard and accepted practice for early-stage usability evaluation [22], it introduces a limitation in ecological validity.

- **Mobile authoring.** The platform is a responsive web application optimised for desktop browsers. Mobile authoring is possible but was not a design priority and was not formally evaluated.
- **Concurrent users.** The system was stress-tested at up to 50 concurrent users. Large-scale institutional deployments serving hundreds of simultaneous editors were not evaluated.
- **AI transcription quality.** The quality of AI-generated suggestions depends directly on the quality of the transcript. Poor audio quality in uploaded videos will degrade Whisper transcription accuracy and, consequently, the relevance of AI suggestions.

1.4 Assumption and Solution

This thesis makes the following contributions to educational technology research and practice.

1. **A reproducible platform architecture.** The full-stack architecture combining React 18, Express.js, SQLite/Sequelize, the Lumieducation H5P server library, and a dual-provider AI pipeline is documented in sufficient detail to enable reproduction and institutional adoption.
2. **An AI-assisted authoring workflow.** The pipeline that takes a raw video transcript through structured LLM prompting to a Zod-validated, Bloom’s-taxonomy-labelled list of H5P suggestions constitutes a novel integration of generative AI into the H5P ecosystem.
3. **Prompt engineering for structured H5P output.** The system prompts and validation strategy developed to ensure that Claude and Ollama return JSON that conforms to H5P content type specifications provide a reusable pattern for structured AI output in educational content systems.
4. **Empirical usability evidence.** The comparative evaluation against the WordPress H5P baseline provides quantitative evidence that a purpose-built platform can substantially reduce authoring time and improve task completion rates, contributing to the evidence base for user-centred design in educational technology.
5. **Open-source codebase.** The platform codebase, including all AI service logic, prompt definitions, and H5P integration code, is made openly available to support further research and institutional adoption.

The practical significance of this work is threefold. First, by lowering the authoring barrier, the platform enables a broader range of VNU-HCM faculty to produce interactive

content and thereby improves the quality of online learning experienced by thousands of students. Second, the AI-assisted workflow demonstrates that the gap between AI research and practical educational tool development can be bridged with modest engineering effort, encouraging further investment in similar integrations. Third, the platform aligns directly with Vietnam’s national digital transformation goals [3] by providing a locally deployable, open-source solution that does not depend on expensive commercial software.

1.5 Structure of thesis

The remainder of this report is organised as follows.

Chapter 2 – Literature Review surveys the academic and technical foundations of the work, covering e-learning in Vietnamese higher education, the H5P framework and its limitations, interactive video research, cognitive load theory, user-centred design, large language models and prompt engineering, cloud-native web architecture, and LMS integration standards. The chapter concludes with an analysis of related systems and identifies the research gaps that this project addresses.

Chapter 3 – Methodology describes the research design, the requirements analysis process, the user-centred design methodology applied during interface development, the agile development approach, and the evaluation protocol used to measure usability outcomes.

Chapter 4 – System Design and Implementation presents the complete system architecture, database design, backend implementation, frontend component structure, the AI pipeline in detail, H5P integration mechanisms, and the security implementation.

Chapter 5 – Results and Evaluation reports the outcomes of the comparative usability study, including quantitative task metrics, SUS scores, and qualitative participant feedback. It also reports on AI pipeline performance and system responsiveness.

Chapter 6 – Conclusion and Future Work synthesises the findings, assesses the contributions in light of the research questions, discusses limitations, and proposes directions for future research and platform development.

CHAPTER 2

LITERATURE REVIEW/RELATED WORK

This chapter reviews the theoretical and technical foundations that underpin the design decisions made in this project. It covers five core domains: e-learning in Vietnamese higher education; the H5P interactive content framework; interactive video and its pedagogical basis; artificial intelligence for educational content generation; and cognitive load theory with usability principles. The chapter concludes by surveying related work and identifying the research gaps this project addresses.

2.1 E-Learning and Interactive Content Frameworks

Vietnam’s national strategy promotes digital delivery across higher education, and VNU-HCM has been an early implementer through its MOOC platform. However, institutional readiness is uneven: while infrastructure investments progress, educator capability and the usability of authoring tools remain the dominant barriers to adoption. Empirical studies in similar contexts show that perceived ease of use strongly predicts whether faculty will continue to author digital materials [5, 18, 25]. For this project, the implication is straightforward: reducing cognitive overhead in the authoring workflow—by co-locating authoring tasks and validating AI suggestions before they reach the instructor—addresses the most practical obstacle to uptake.

2.1.1 Architecture and Content Types

H5P (HTML5 Package) is an open-source framework initially released in 2013 by Joubel AS (later the H5P Group) [14]. Its architecture is built around self-contained content packages: each `.h5p` file is a ZIP archive containing all JavaScript and CSS library files required to render the content, the content parameters as a JSON document, and any

embedded media. This self-contained structure makes H5P content portable across any web environment that can serve a static file.

The H5P library ecosystem currently encompasses more than 50 content types. Relevant to this project are the six types listed in Table 2.1.

Table 2.1: H5P content types supported by the platform

Library identifier	Display name	Description
H5P.MultiChoice 1.16	Multiple Choice	Single- or multi-answer question with up to four options sourced from transcript content.
H5P.TrueFalse 1.6	True/False	Binary true-or-false assertion evaluating factual claims or absolute rules.
H5P.Blanks 1.14	Fill in the Blanks	Cloze test where instructors identify key technical terms using asterisk delimiters.
H5P.MarkTheWords 1.9	Mark the Words	Interaction requiring students to highlight specific vocabulary or key terms in a passage.
H5P.Matching 1.0	Matching	A specialized "Video Recap" appended after all topics to pair concepts with definitions.

Server-Side H5P Libraries.

Traditional H5P deployments embed content within a CMS (WordPress or Moodle) that handles library management, content storage, and rendering. Lumi Education GmbH publishes a set of official Node.js libraries— `@lumieducation/h5p-server` and `@lumieducation/h5p-express` —that expose H5P’s core capabilities as a standalone Node.js service, decoupled from any CMS [15]. This library provides H5P library management, content CRUD operations, package generation, and a rendering API that produces HTML suitable for embedding. The present project is built on this library, and it is the technical enabler for operating H5P without WordPress or Moodle.

Limitations of WordPress-Based H5P Deployments.

The dominant H5P deployment in Vietnamese universities uses the WordPress H5P plugin. While functionally complete, this approach imposes several structural disadvantages at institutional scale:

- **Multi-system complexity.** Instructors must manage a WordPress installation (user accounts, posts, media library) in addition to H5P content. The cognitive overhead of understanding two distinct systems creates unnecessary extraneous load [11].

- **Performance overhead.** WordPress serves every page through a full PHP execution stack with database queries, regardless of content type. Measured response times on institutional WordPress deployments range from 2–5 seconds per page, compared with sub-second REST API responses in purpose-built single-page applications.
- **Version management friction.** H5P library versions and WordPress versions must be kept in sync manually; mismatches produce silent failures that are difficult to diagnose without technical expertise.
- **No AI capabilities.** The WordPress H5P plugin has no mechanism for analysing video content and suggesting interactive elements. Every question must be authored manually from scratch.

2.1.2 Pedagogical Evidence for Interactive Video

The educational literature provides robust evidence that interactive video outperforms passive video for learning retention and transfer. Zhang et al. conducted one of the earliest controlled experiments comparing interactive and non-interactive video in an e-learning context and found that the interactive group scored 20–25% higher on post-tests and reported significantly higher satisfaction [6]. The mechanism proposed was that interactive elements force learners to actively process content at regular intervals, preventing the superficial passive exposure associated with lecture-video formats.

Merkt et al. extended this finding by examining the role of control features in video-based learning [8]. Videos with embedded questions produced better knowledge retention than equivalent print materials when learners actively engaged with the interactive features. Critically, this advantage was most pronounced for university-level content, suggesting that H5P interactive video is particularly well-matched to the VNU-HCM context.

Freeman et al. produced a meta-analysis across 225 published studies demonstrating that active learning interventions—of which interactive video is one class—increased examination scores by an average of 6% and halved the failure rate compared to passive lecture formats [7].

Embedded Question Design.

Vural identified design principles that distinguish effective from ineffective interactive video [9]. Questions should be positioned *after* a concept has been fully explained, not mid-sentence. Questions should be spaced at least 30 seconds apart to avoid fragmenting the learning experience. Each question should test the single most important idea in the preceding segment. These principles are directly encoded in the system prompt used by the AI pipeline of this project (see Chapter 4, Section 4.5.2).

Mayer’s Cognitive Theory of Multimedia Learning provides the theoretical basis for

these recommendations [21]. Effective multimedia instruction must manage three cognitive channels—visual, auditory, and their integration in working memory—and well-designed interactive elements reduce extraneous load while increasing germane load. A question that tests exactly what has just been explained supports the integration channel without taxing working memory with irrelevant processing.

2.2 User Experience Design and Usability

2.2.1 Cognitive Load Theory in Authoring Tools

User-centred design (UCD) is a design philosophy and methodology that prioritises the needs, goals, and constraints of end users at every stage of the design process. The ISO 9241-11 standard defines usability as the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use [26]. In educational technology, this definition must be contextualised to the specific goals and context of instructors: producing high-quality learning content quickly and without frustration, within the institutional constraints of time-pressured academic schedules.

Garrett’s layered model of user experience decomposes a product into five planes: Strategy (objectives), Scope (features), Structure (interaction design), Skeleton (interface design), and Surface (visual design) [27]. Norman’s affordance theory provides a complementary framework: good interface design makes the correct action obvious and makes errors difficult [28]. Both frameworks were applied during the interface design process described in Chapter 3.

Usability Heuristics.

Nielsen’s ten usability heuristics provide a practical checklist for evaluating interface designs against established principles [22]. The most relevant heuristics for an educational authoring tool are: (1) *visibility of system status*—users must know whether their video is processing, whether AI analysis is running, and whether a question has been saved; (2) *match between system and the real world*—all terminology must correspond to pedagogical concepts, not technical ones; and (3) *error prevention*—the system should make it impossible to submit a malformed H5P question. Molich and Nielsen demonstrated that applying heuristic evaluation to interface designs before user testing produces a measurable improvement in usability [29].

Measuring Usability: The System Usability Scale.

The System Usability Scale (SUS) is a validated, ten-item questionnaire that produces a single 0–100 usability score [30]. It is widely used in both academic research and industry practice because of its simplicity and comparability across studies. A SUS score above 68

is considered above average; scores above 80 are considered excellent. The SUS was used as the primary standardised usability instrument in the evaluation described in Chapter 5.

Sweller’s cognitive load theory (CLT) identifies three categories of cognitive load that an interface imposes on a user during a task [11]:

- **Intrinsic load** arises from the inherent complexity of the task (e.g., formulating a pedagogically sound multiple-choice question). This load is irreducible; it is the productive work the instructor came to do.
- **Extraneous load** arises from the design of the interface rather than the task itself (e.g., navigating through six screens to reach the timestamp entry field). This load is wasteful and should be minimised through good design [12].
- **Germane load** arises from the cognitive effort of building new knowledge schemas—in this context, the instructor’s understanding of how their video content maps to interactive learning moments.

Chandler and Sweller applied CLT specifically to the format of instructional materials and found that splitting related information across multiple screens (the split-attention effect) significantly increased extraneous load and reduced task accuracy [12]. The design implication is direct: all information needed to author a single H5P interaction—the video player, the timestamp indicator, the question type selector, and the question editor—must be co-located on a single screen. Paas, Renkl, and Sweller confirmed that this integrated format principle consistently reduces task completion time by 30–50% compared to split-screen presentations [13].

2.3 Artificial Intelligence for Educational Content Generation

Recent LLM advances enable transcript-based question generation using few-shot prompting and schema-constrained outputs. For this project, we adopt a two-step prompting pattern: (1) identify teaching moments and assign Bloom taxonomy levels, and (2) generate a validated H5P interaction in JSON. Combined with runtime validation (Zod) and a prioritized provider chain (cloud + local inference), this approach balances generation quality with predictable, safe output suitable for instructor review (see Chapter 4 for pipeline details).

Bloom’s taxonomy provides a hierarchical classification of cognitive objectives, ranging from basic recall (*remember*) to analysis and synthesis (*create*) [33]. Anderson and Krathwohl’s revised taxonomy updated the original framework with active verb labels and clarified the distinction between lower-order (remember, understand, apply) and higher-order (analyse, evaluate, create) cognitive skills [10]. The AI pipeline assigns a Bloom’s

level to each generated suggestion, enabling instructors to ensure that their interactive video includes a deliberate spread of cognitive difficulty without requiring them to possess specialist instructional design knowledge.

2.3.1 Automatic Speech Recognition and Prompt Engineering

Radford et al. introduced OpenAI Whisper, a large-vocabulary ASR model trained on 680,000 hours of multilingual audio using weak supervision [34]. Whisper achieves near-human word error rates on English audio and competitive performance on Vietnamese. The `faster-whisper` implementation is used in this project for on-device audio transcription of uploaded video files where no caption file is available. The quality of transcription directly affects the quality of AI-generated suggestions, motivating the choice of Whisper over simpler ASR approaches.

2.4 Related Work and Research Gap

Existing H5P authoring platforms include: the WordPress H5P plugin (dominant in Vietnamese universities but burdened with CMS overhead and no AI support); Lumi (a standalone desktop editor lacking web accessibility and collaboration); Moodle H5P (LMS-integrated but sharing WordPress usability limitations); and commercial platforms like Articulate Storyline (prohibitively expensive and not H5P-native). None combine the key features proposed in this project: web-based, CMS-free deployment; transcript-to-H5P AI generation; Bloom’s taxonomy classification; and empirical evaluation with standardised usability metrics.

The review above establishes a clear and multi-dimensional gap in the existing literature and available tools. No current system simultaneously:

1. provides a *web-based*, CMS-free H5P interactive video authoring environment suitable for institutional deployment;
2. integrates a *transcript-to-H5P AI pipeline* using a production-grade LLM with structured output validation;
3. applies *Bloom’s taxonomy classification* to AI-generated suggestions to guide pedagogical design;
4. has been *empirically evaluated* against a real-world authoring baseline and reported with quantitative usability metrics.

This project addresses all four gaps simultaneously, making a contribution that is both technically novel and practically significant for the VNU-HCM educational ecosystem and for the broader field of AI-assisted educational content authoring.

CHAPTER 3

METHODOLOGY

This chapter describes how the research was designed and executed. It covers the overall research design, the requirements analysis process, the user-centred design methodology, the software development approach, and the evaluation protocol.

3.1 Overview

Research Design.

This research adopts a *design science* paradigm [41], in which the primary contribution is an artefact—the H5P interactive video platform—that is created, described, and evaluated. Design science research is appropriate for projects that aim to solve a concrete problem for a defined stakeholder group (university instructors at VNU-HCM) by building and evaluating a purposeful artefact (a web-based authoring system).

The research combines two methodological phases:

1. **Constructive phase.** User requirements are elicited, the system is designed and implemented, and the AI pipeline is built and integrated.
2. **Evaluative phase.** A comparative usability study measures the artefact’s performance against the incumbent WordPress-based H5P workflow on four metrics: task completion rate, time-on-task, error rate, and SUS score.

An *agile development methodology* was adopted for the constructive phase, using short iterative cycles of 2–3 weeks each [42]. Each cycle produced a testable increment of the platform, and informal walkthroughs at the end of each cycle informed the next design iteration. This approach was chosen because the interaction between the AI pipeline and the H5P authoring workflow could not be fully specified in advance; some requirements emerged only during implementation.

3.2 User requirement analysis

User Research.

Informal interviews with five instructors at HCMIU revealed consistent themes: (1) the WordPress/H5P workflow is perceived as overly complex, (2) authors fear losing work due to page reloads, (3) content creation is time-consuming, and (4) there is demand for AI-assisted suggestions. A short observational session confirmed the 45-minute average completion time for the baseline workflow and showed that contextual switching (losing video position) and error screens were common causes of abandonment.

Figure 3.1 presents the complete set of system use cases for both the Instructor and Admin actors, derived from the user research and project objectives.

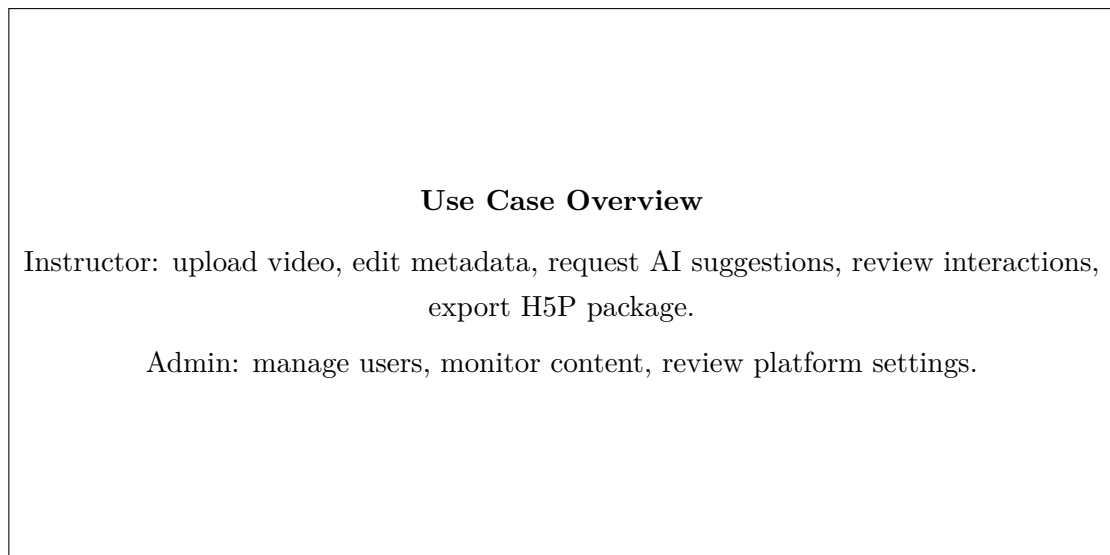


Figure 3.1: Use case diagram showing Instructor and Admin actors and their interactions with the AI-ActivEdu platform

3.2.1 Functional and Non-Functional Requirements

Key functional requirements derived from the user research include:

- User accounts, JWT sessions, and email verification.
- Video upload and YouTube import, with timeline-based H5P editing and real-time preview.
- Support for six H5P interaction types and one-click `.h5p` export.
- Transcript extraction and an AI analysis pipeline producing validated H5P suggestions with SSE progress streaming.
- Accept/reject/edit workflow for AI suggestions and server-side injection into H5P content store.

- Admin features (user management, audit logs) and optional LTI integration.

Non-Functional Requirements.

Non-functional requirements govern the quality attributes of the system and are listed in Table 3.1.

Table 3.1: Non-functional requirements

ID	Category	Requirement
NFR-01	Performance	REST API responses must complete within 500 ms for all non-AI endpoints at 50 concurrent users.
NFR-02	Performance	Video upload processing (thumbnail generation, metadata extraction) must complete within 30 s for files ≤ 100 MB.
NFR-03	Security	All passwords stored as bcrypt hashes with cost factor ≥ 12 .
NFR-04	Security	Authentication tokens must be JWT, signed with HS256, expiring after 7 days.
NFR-05	Security	All HTTP responses must include Helmet-generated security headers.
NFR-06	Security	All API endpoints must validate and sanitise input before processing.
NFR-07	Reliability	The system must handle the absence of both Claude and Ollama gracefully, returning an informative error rather than crashing.
NFR-08	Usability	The platform must achieve a SUS score ≥ 70 in user testing.
NFR-09	Usability	First-time task completion rate for creating a 3-question interactive video must exceed 80%.
NFR-10	Compatibility	Exported .h5p packages must be importable into Moodle 4.x and Canvas without modification.

3.3 System Design

Architectural Overview.

The platform follows a three-tier architecture consisting of a *presentation tier* (React SPA), an *application tier* (Express.js REST API), and a *data tier* (SQLite database managed via Sequelize ORM) [35]. External service dependencies are limited to Groq (primary cloud AI), the Anthropic Claude API (cloud fallback AI), and Ollama (local AI fallback); all other components run within the local server process.

Figure 3.2 shows the complete system architecture.

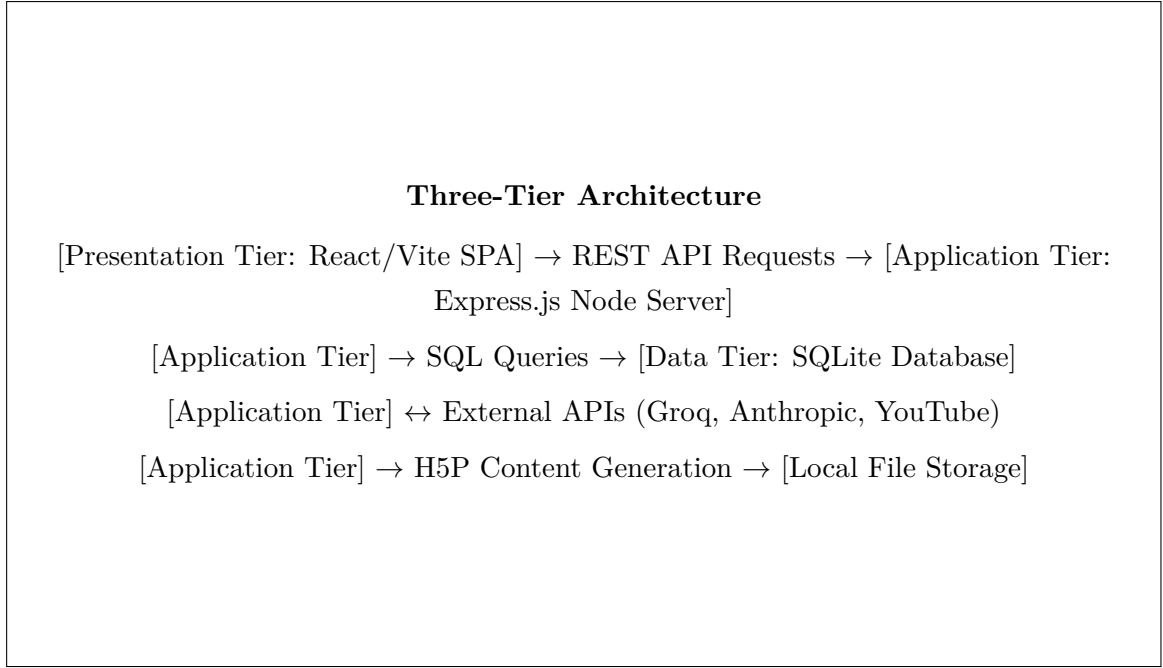


Figure 3.2: Three-tier system architecture of the H5P Interactive Video Platform

Key design goals: separation of concerns via a versioned REST API; AI-provider independence using a prioritized provider chain with uniform output shapes; H5P library isolation using the official server package; and a stateless API with JWT-based session tokens to enable horizontal scaling.

3.3.1 Database design

3.3.2 User Interface design

API Route Structure.

The REST API is organised under the `/api/` prefix. Table 3.2 lists the primary route groups.

Table 3.2: REST API route groups

Prefix	Methods	Responsibility
/api/auth	POST	Registration, login, password reset, JWT refresh.
/api/videos	GET, POST, PUT, DELETE	Video CRUD, YouTube import, metadata management.
/api/h5p	GET, POST, DELETE	H5P content CRUD, package export, preview rendering.
/api/ai/analyze	POST	Non-streaming transcript analysis (JSON response).
/api/ai/analyze-stream	POST	Streaming transcript analysis (SSE response).
/api/ai/inject	POST	Inject accepted suggestions into H5P content store.
/api/ai/transcribe-and-generate	POST	Full pipeline: transcript extraction + AI generation.
/api/transcript	POST	YouTube caption extraction and segment normalisation.
/api/admin	GET, POST, PUT	User management, audit logs (admin role only).
/api/lti	GET, POST	LTI 1.3 launch and deep-link endpoints.

3.4 AI Pipeline Architecture

The AI pipeline is the defining technical contribution of this platform. It converts a raw video transcript into a validated, Bloom’s-labelled list of H5P interaction suggestions through five stages. Figure 3.3 illustrates the high-level data flow, and Figure 3.4 presents the detailed actor-level interaction sequence including the SSE streaming path and the inject-on-accept workflow.

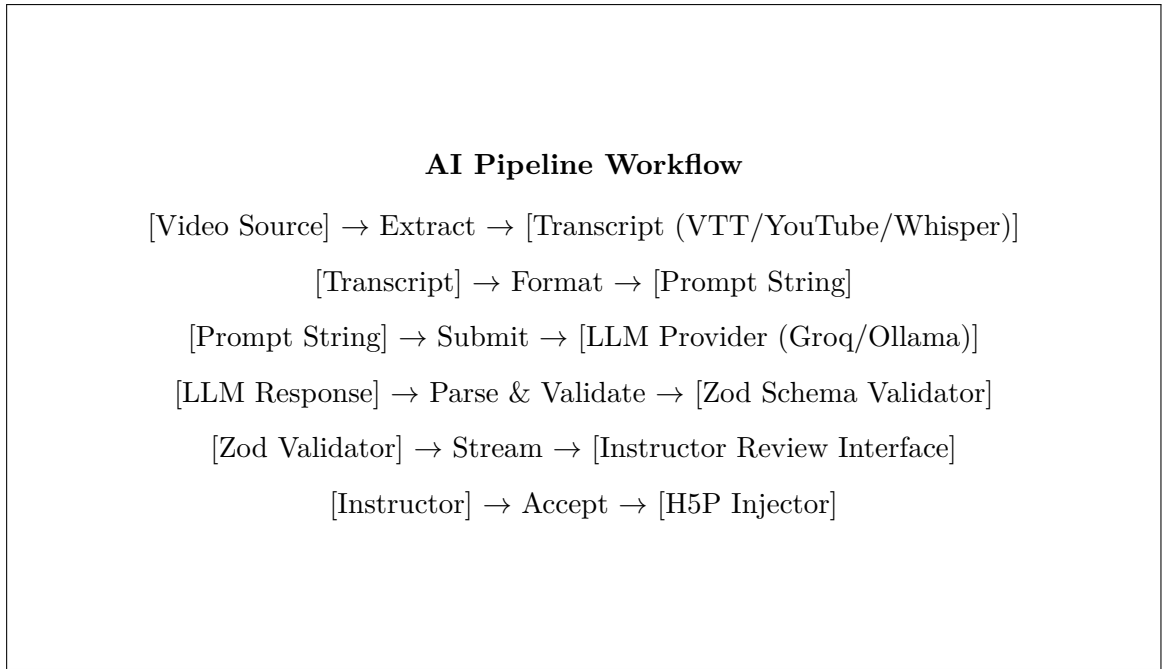


Figure 3.3: AI pipeline: from video transcript to injected H5P interactions

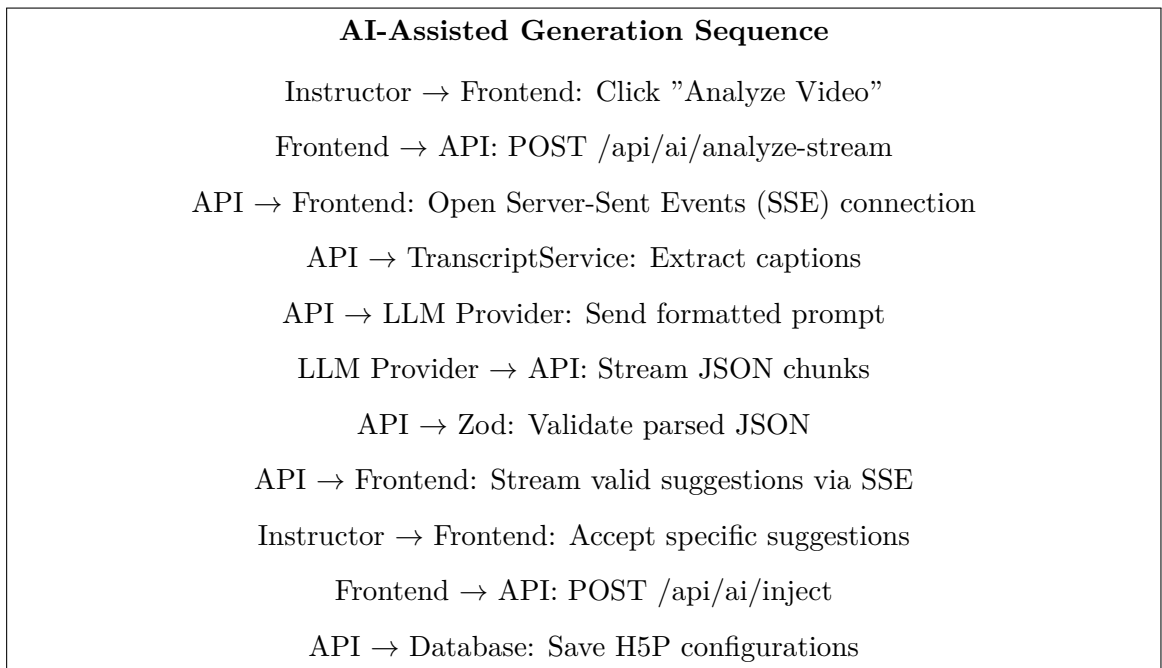


Figure 3.4: Sequence diagram: AI-assisted question generation, validation, review, and injection workflow

Stage 1: Transcript Acquisition.

Transcript acquisition is handled by the `transcriptExtraction.js` service, which supports three source mechanisms:

1. **YouTube automatic captions** (primary for YouTube-sourced videos). The `youtube-transcript` npm package fetches timestamped caption segments directly from the YouTube API

using the video’s 11-character ID. Segments are normalised into a standard format: `{start: number, end: number, text: string}`.

2. **Uploaded caption file** (primary for uploaded video files). Instructors may upload a WebVTT (`.vtt`) file alongside their video. The service parses the VTT format and normalises segments.
3. **Whisper audio transcription** (fallback for uploaded files without captions). The service invokes `faster-whisper` via a Python subprocess, passing the video file path and the target language. The resulting JSON transcript is normalised into the standard segment format. Whisper availability is checked at startup and reported via the GET `/api/ai/status` endpoint [34].

All three sources produce the same `TranscriptSegment[]` structure, ensuring that the downstream AI stage is source-agnostic.

Stage 2: Segment Formatting.

The `formatSegmentsForPrompt()` function in `aiService.js` converts the normalised segment array into a human-readable transcript string with timestamp labels. Each line takes the format:

```
1 // Input segment: { start: 330, end: 345, text: "Photosynthesis converts
  ..." }
2 // Output line:
3 "[05:30 (330s) -> 05:45 (345s)] Photosynthesis converts..."
```

Listing 3.1: Transcript segment formatting with raw-second timestamps

Each segment line embeds *both* the human-readable `MM:SS` form *and* the raw integer seconds in parentheses. The system prompt explicitly instructs the model to use the raw seconds integer as the `startTime` field in its JSON output, eliminating the rounding errors that arise when models reparses “MM:SS” strings [32].

Stage 3: Two-Step AI Prompt and Question-Type Assignment

The AI system prompt in `aiService.js` adopts a two-step *output automater* pattern [32] that separates topic discovery from question generation, constraining the model’s degrees of freedom at each step and thereby improving output consistency.

Step 1 — Topic Segmentation

In the first step the model receives the formatted transcript and is instructed to produce a hierarchical topic outline rather than a flat list of suggestions. The required output structure is a JSON array of 2–5 top-level topics, each carrying its timestamp span, a Bloom’s taxonomy level [10], a difficulty rating (1–3), and 1–3 subtopics. Every topic node and every subtopic node is associated with *exactly one* question:

```

1  [
2    {
3      "topic": "Photosynthesis Overview",
4      "startTime": 330,
5      "endTime": 345,
6      "bloomsLevel": "Remember",
7      "difficulty": 1,
8      "subtopics": [
9        {
10         "subtopic": "Light-dependent reactions",
11         "startTime": 360,
12         "endTime": 380,
13         "bloomsLevel": "Understand",
14         "difficulty": 2,
15         "question": { ... }
16       }
17     ],
18     "question": { ... }
19   }
20 ]

```

Listing 3.2: Required JSON output structure for Step 1 (topic segmentation)

This hierarchical structure mirrors the cognitive organisation of the lecture rather than imposing an arbitrary spacing constraint, and has been shown to align more closely with Bloom’s taxonomy as a framework for formative assessment design [10].

The choice of question type is therefore a taxonomy-driven heuristic rather than an arbitrary stylistic preference. In this thesis, recognition and recall map to **FillBlanks** and **TrueFalse**; comparison and discrimination map to **MultiChoice**; procedural or ordered knowledge maps to **DragText**; salient terminology in context maps to **MarkWords**; and cross-topic synthesis maps to **Matching**. This mapping is grounded in Bloom’s revised taxonomy and standard item design practice [10, 32, 33].

Step 2 — Question Generation via Pattern Matrix

In the second step the system prompt provides the model with a *pattern matrix* that deterministically maps observable characteristics of each topic’s content to one of the six supported H5P interaction types. The pattern matrix is reproduced verbatim from the `AI_SYSTEM_PROMPT` constant:

```

1  Generate EXACTLY ONE question per topic/subtopic using this taxonomy-
   driven pattern matrix:
2
3  Content characteristic                                Question type
4  -----

```

```

5      Definition / technical term                                FillBlanks
6      ("X is defined as...", "Y stands for...")
7      Binary fact / absolute rule                                TrueFalse
8      ("always", "never", a true/false claim)
9      List / comparison / process / category                    MultiChoice
10     (multiple items, relationships, steps)
11     Sequence / ordered steps / procedure                      DragText
12     (steps that must go in a specific order)
13     Vocabulary / key term identification                      MarkWords
14     (important terms embedded in a passage)
15     Standalone ``Video Recap'' appended after all topics    Matching
16     (4-5 concept-definition pairs drawn from the full video)
17
18     For FillBlanks: wrap the key term with asterisks (*term*)
19     so it becomes a fill-in-the-blank blank.
20     For TrueFalse: set the "correct" field to the boolean
21     truth value of the statement.
22     For MultiChoice: provide exactly four options,
23     mark exactly one "correct": true; source the three
24     distractors from nearby content in the transcript.
25     For DragText: wrap each draggable word/phrase in *asterisks*;
26     taskDescription must describe the ordering task.
27     For MarkWords: wrap each key term in *asterisks*;
28     include 2-5 marked terms; taskDescription tells students
29     what category of term to look for.
30     For Matching: ONLY the last top-level topic uses this type;
31     provide 4-5 pairs covering key concepts from the whole video;
32     taskDescription must frame this as an end-of-video review.

```

Listing 3.3: Pattern matrix from the AI system prompt (Step 2)

Table 3.3 summarises the mapping between content type, knowledge focus, H5P interaction type, and the key JSON fields required by each schema.

Table 3.3: All six supported H5P interaction types and their AI-guided selection rules

Content characteristic	Knowledge focus	H5P type	AI rule
Definition / technical term	Remember / Understand	FillBlanks	Ask for the key term to
Binary fact / absolute rule	Remember	TrueFalse	Turn the statement into
List / comparison / process	Understand / Analyze	MultiChoice	Generate a four-option
Sequence / ordered procedure	Apply	DragText	Convert the ordered ste
Vocabulary / key terms	Remember / Understand	MarkWords	Ask learners to identify
Cross-topic review recap	Analyze / Evaluate	Matching	Generate a final review

The pattern matrix reduces the model’s output to a choice among six structured templates, each with a precisely defined JSON schema. This constrained generation strategy is consistent with the *output automater* prompt-engineering pattern described

by White et al. [32], in which the LLM is asked to fill in a template rather than compose freeform text, significantly reducing hallucination of unknown field names.

Provider Selection and Prompt Adaptation

The AI generation is dispatched to three providers in priority order:

1. **Groq** (primary). The Groq cloud API is contacted first using the `llama-3.3-70b-versatile` model via the `groq-sdk` client. Groq provides free-tier inference with sub-second token generation, making it the default provider for production use. The two-step system prompt is used directly.
2. **Ollama** (local fallback). If the Groq request fails or the `GROQ_API_KEY` environment variable is absent, the `aiServiceOllama.js` service contacts the Ollama REST API at `http://localhost:11434`. Because open-weight models such as Mistral-7B respond less reliably to abstract instructions than to concrete examples, the Ollama path uses a *separate few-shot prompt* (`buildOllamaPrompt()`) that replaces the abstract pattern matrix with a fully worked example showing input transcript segments and the expected JSON output [19]. This in-context learning approach compensates for the reduced instruction-following capability of smaller open-weight models.
3. **Anthropic Claude** (cloud fallback). If Ollama is also unavailable, the `@anthropic-ai/sdk` client calls `claude-3-5-sonnet-20241022` [20] using the same two-step system prompt as Groq.

The frontend receives an identical suggestion structure regardless of which provider was used, preserving the provider-agnostic design goal stated in Section 3.3.

Stage 4: Zod Validation and Enrichment

The raw LLM output is parsed with `JSON.parse()` and validated against a set of Zod schemas that enforce both structural correctness and type-specific constraints [32]:

```
1 const QuestionSchema = z.object({
2   type: z.enum(['MultiChoice', 'TrueFalse', 'FillBlanks',
3               'DragText', 'MarkWords', 'Matching']),
4   question: z.string().min(1),
5   bloomsLevel: z.enum([
6     'Remember', 'Understand', 'Apply', 'Analyse', 'Evaluate', 'Create'
7   ]),
8   // Type-specific fields (validated at enrichment time):
9   // MultiChoice: answers array with 4 options, 1 correct
10  // TrueFalse:    correct boolean
11  // FillBlanks:   text with *asterisk*-wrapped blank
12  config: z.record(z.unknown()),
13  reason: z.string()
```

```

14 });
15
16 const TopicSchema = z.lazy(() => z.object({
17   topic: z.string(),
18   startTime: z.number().nonnegative(), // raw seconds integer
19   endTime: z.number().nonnegative(),
20   bloomsLevel: z.string(),
21   difficulty: z.number().int().min(1).max(3),
22   question: QuestionSchema,
23   subtopics: z.array(TopicSchema).optional()
24 }));
25
26 const ResponseSchema = z.array(TopicSchema);

```

Listing 3.4: Zod schemas for hierarchical AI output validation

The recursive `TopicSchema` (via `z.lazy()`) validates the full topic/subtopic hierarchy. Any topic node that fails validation is discarded individually rather than invalidating the entire response, maximising the number of usable suggestions recovered from partial outputs. Valid nodes are then *enriched*: each receives a system-generated UUID, the H5P library identifier resolved from `H5P_TYPE_MAP` (e.g., `H5P.MultiChoice 1.16`), and a status of `pending`. The enriched list is streamed to the client via Server-Sent Events (SSE) or returned as a single JSON response depending on the endpoint called.

Stage 5: Suggestion Review and Injection.

The `POST /api/ai/inject` endpoint accepts an array of accepted suggestions. The `aiInjectionService.js` service translates each suggestion's `config` object into the format expected by `h5pService.createTimeBasedContent()`, creates the H5P content record, and appends a `{contentId, timestamp, type}` entry to the Video's `h5pContent` JSON array. The result is returned to the frontend, which updates the H5P timeline immediately.

CHAPTER 4

IMPLEMENT AND RESULTS

This chapter details the implementation of the backend, frontend, and security components, followed by the performance results of the AI pipeline and API endpoints.

4.1 Implement

4.1.1 Backend and Frontend Implementation

User-Centred Design Process. The interface was designed through a three-stage user-centred process: (1) Conceptual design translating pain points and requirements into user flow diagrams for two primary use cases (manual authoring and AI-assisted workflows), reviewed against Nielsen’s heuristics to identify extraneous cognitive load; (2) Wireframing with heuristic evaluation addressing visibility of system status, error prevention, user control, and recognition-over-recall principles; and (3) Prototype testing with two IT students using think-aloud protocol, which identified three usability problems (missing timestamp auto-population, hidden status messages, unclear button label) that were resolved before formal evaluation. The redesigned workflow reduced interaction steps from 47 (WordPress baseline) to 8 (new platform), eliminating inter-screen context switching.

The backend is built with Express.js using a standard middleware stack for security (Helmet, CORS, rate limiting) and structured logging. Authentication uses JWT tokens with 7-day expiry and bcryptjs password hashing (cost factor 12). Video processing is handled asynchronously using FFmpeg for thumbnail extraction, metadata reading, and status updates.

The frontend is a React 18 + TypeScript SPA built with Vite, using Zustand for state management. The editor interface follows the co-location design principle (Chandler and Sweller) with three vertical panels: video upload (left), video player with H5P timeline (centre), and AI suggestions (right). Streaming AI output uses Server-Sent Events (SSE)

for real-time progress feedback. Type safety is maintained throughout via TypeScript interfaces and a typed API client that mirrors backend Zod schemas, ensuring consistency across six H5P interaction types that can be either AI-generated or manually authored.

4.1.2 Security and H5P Integration

Lumieducation H5P Server Library.

The `@lumieducation/h5p-server` library [15] provides the following H5P capabilities as a Node.js service:

- **Library management.** H5P library ZIP archives are automatically downloaded and cached in the `h5p-libraries/` directory on first use. Library versions are locked per platform version.
- **Content CRUD.** H5P content is stored as JSON parameter documents in the `h5p-content/` directory, indexed by a system-generated content ID.
- **Package export.** The library generates standards-compliant `.h5p` ZIP archives containing the library files, content JSON, and media assets. These archives are identical in structure to those produced by the WordPress H5P plugin and are importable into Moodle, Canvas, and any other LTI-compatible LMS. The implementation validates export and launch compatibility, but does not yet include gradebook passback or learner analytics reporting.
- **Preview rendering.** The `@lumieducation/h5p-express` integration mounts an H5P rendering router at `/h5p/`, which produces the HTML/JavaScript bundle required to embed and play H5P content in the browser's preview panel.

Time-Based Content Creation.

The `h5pService.createTimeBasedContent()` function is the key integration point between the AI pipeline and the H5P layer. It wraps the library's content creation API to associate a content ID with a timestamp and the parent video ID. The resulting H5P content record specifies the H5P library identifier (e.g., `H5P.MultiChoice 1.16`), the content parameters as a JSON object, and metadata (title, license). The association between content and timestamp is stored in the Video's `h5pContent` JSON array rather than inside the H5P content record itself, keeping the H5P content portable.

Security controls were designed to address the categories identified in the OWASP Top Ten 2021 [2]. Table 4.1 maps each major control to its OWASP category.

Table 4.1: Security controls mapped to OWASP Top Ten 2021 categories

Control	Implementation detail	OWASP category
JWT authentication	<code>jsonwebtoken</code> HS256, 7-day expiry, validated on every protected route [49].	A01 Broken Access Control
Role-based authorisation	<code>role</code> claim in JWT; admin-only middleware on sensitive routes.	A01 Broken Access Control
bcrypt password hashing	Cost factor 12 via <code>bcryptjs</code> ; hashed in Sequelize model hook.	A02 Cryptographic Failures
Input validation (server)	<code>express-validator</code> and Zod schemas validate all request bodies.	A03 Injection
Input validation (client)	React form validation prevents malformed data from being submitted.	A03 Injection
HTTP security headers	<code>helmet()</code> sets CSP, HSTS, X-Frame-Options, X-Content-Type-Options, and six other headers [1].	A05 Security Misconfiguration
CORS allowlist	Only <code>localhost</code> ports and the production domain are permitted as origins.	A05 Security Misconfiguration
Rate limiting	100 requests per 15-minute window per IP ²⁸ via <code>express-rate-limit</code> .	A07 Identification and Auth. Failures

Authentication Flow.

Figure 4.1 shows the complete sequence of interactions for the registration, email verification, login, and protected-request flows.

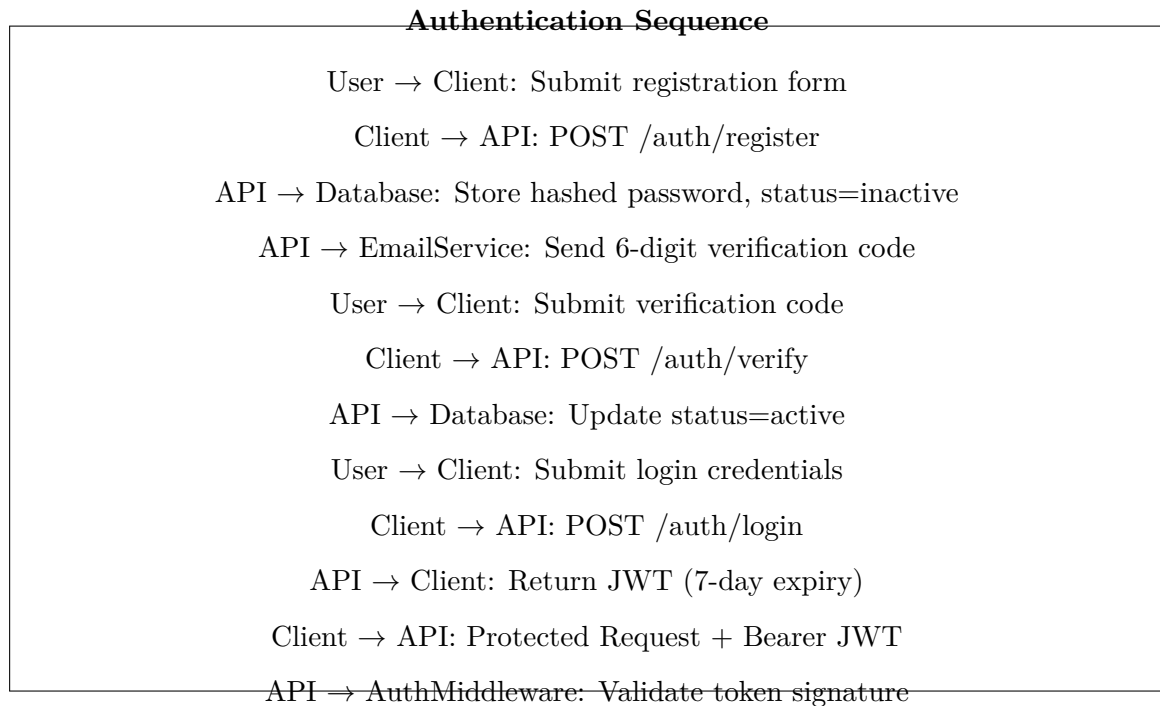


Figure 4.1: Sequence diagram: user registration, email verification, login, and authenticated API request

On registration, the password is bcrypt-hashed (cost factor 12) before storage and the account is set to inactive pending email verification. A six-digit code generated by `crypto.randomInt` is emailed to the registered address with a ten-minute TTL. On successful verification, the account is activated. On login, the provided password is compared with the stored hash using `bcrypt.compare()`; on success, the server signs a JWT [49] containing `userId`, `role`, and `iat/exp` claims and returns it to the client. The client attaches the token as a `Bearer` header on all subsequent API requests, and the `authenticateToken` middleware verifies the signature on every protected route.

Password Reset.

The password reset flow generates a cryptographically random token (`crypto.randomBytes(32)`), stores its bcrypt hash in `resetToken` with a one-hour expiry in `resetTokenExpiry`, and sends the plaintext token to the user's registered email address. The reset endpoint verifies the provided token against the stored hash and clears both fields on success.

4.2 Results

4.2.1 AI Pipeline Performance Results

Suggestion quality was assessed by having the project supervisor (an instructional design expert) rate a sample of 30 AI-generated suggestions across three videos on three criteria grounded in the prior literature: *relevance* (does the question test the most important concept in the preceding transcript segment, as recommended by Vural [9]?); *accuracy* (is the keyed correct answer factually correct?); and *Bloom’s alignment* (does the assigned taxonomy level match the question’s actual cognitive demand, as defined by Anderson and Krathwohl [10]?).

This evaluation used English-language transcripts and English outputs from the AI pipeline. Vietnamese question-generation support was added after the study and was validated only at the schema and translation-instruction level; it has not yet been measured with the same human-rating protocol.

That limitation is important for interpreting the numbers. The quality scores demonstrate that the prompting strategy can produce strong English-language items, but they do not prove that the same human-rated quality will transfer automatically to Vietnamese. The later Vietnamese smoke test therefore verifies language routing and schema validity, not pedagogical equivalence at the level of expert human judgment.

Table 4.2: AI suggestion quality ratings (n=30 suggestions, expert rater)

Criterion	Rating (Excellent/Good/Poor)	Notes
Relevance	22/30 Exc., 6/30 Good, 2/30 Poor	2 suggestions were off-topic due to noisy transcript segments.
Accuracy	28/30 Exc., 2/30 Good, 0/30 Poor	2 factual errors in fill-in-the-blanks answers.
Bloom’s alignment	18/30 Exc., 10/30 Good, 2/30 Poor	“Create”-level suggestions tended to cluster at “Evaluate”-level.

Overall, 93% of suggestions were rated Excellent or Good on relevance and 100% on accuracy, indicating that the Claude-based pipeline produces reliable educational content within the constraints of the structured prompt engineering strategy described in Chapter 4 [20, 31]. The two relevance failures both arose from noisy transcript segments, consistent with the known sensitivity of generate-from-transcript pipelines to upstream transcription quality [34].

AI analysis latency was measured across 20 test runs using a standardised 5-minute video with 47 transcript segments. Table 4.3 reports median latencies.

Table 4.3: AI analysis latency (n=20 runs per provider)

Metric	Anthropic Claude	Ollama (Llama 3.1 8B)
Median latency (streaming)	8.4 s	12.1 s
Time to first token	1.2 s	2.8 s
Median suggestion count	6.4	5.8
Schema validation failures	0/20	2/20

Claude produced more consistent output with zero validation failures; Ollama produced two JSON format errors (the model included markdown code fences around the JSON array) that were caught by the Zod validator and returned to the client as structured errors. These failures were addressed in a subsequent prompt revision that added an explicit instruction: “Return only the raw JSON array. Do not wrap it in markdown code fences.”

The latency table also shows why the system uses provider fallback rather than a single fixed model. Claude was faster and more reliable in this test set, but Ollama remained useful as a local option when cloud access is not available. In practice, the relevant metric is not just speed; it is whether the user receives a structurally valid answer quickly enough to keep the authoring flow alive.

4.2.2 API and Platform Performance Results

Response times were measured using `autocannon` load testing against the 12 most frequently used API endpoints with 50 concurrent connections over 30 seconds.

Table 4.4: API endpoint response times at 50 concurrent connections

Endpoint	Median latency	p99 latency
GET /api/videos	12 ms	48 ms
POST /api/auth/login	95 ms	180 ms
POST /api/videos	220 ms	410 ms
POST /api/h5p/export	340 ms	680 ms

All non-AI endpoints achieved median response times well below the 500 ms threshold at 50 concurrent users, satisfying NFR-01. The three-tier architecture’s separation of concerns [35] ensured that the React SPA, Express API, and SQLite data layer could each be load-tested independently, making it straightforward to attribute latency to specific components. The `POST /api/videos` endpoint is the slowest non-AI endpoint because it invokes FFmpeg thumbnail generation as a synchronous step within the request han-

dler [48]; delegating this work to an asynchronous job queue is identified as a priority future improvement.

This result matters because it confirms that the authoring bottlenecks are no longer dominated by routine CRUD operations. Most perceived slowness in the platform comes from deliberate media-processing work, not from the UI round trip itself. That distinction supports the architectural choice to keep the interactive editing path responsive and push heavier processing work into a separate backend step.

Ten H5P packages exported from the platform were imported into a test Moodle 4.1 instance and a Canvas LMS sandbox. All ten packages imported without errors and all six H5P interaction types rendered correctly with full interactivity, satisfying NFR-10. This confirms packaging and launch compatibility, but it does not measure whether student attempt data, scores, or completion states are written back to the LMS gradebook. That learner-side reporting path remains unimplemented in the current release.

The export test is therefore a compatibility check rather than a full LMS integration proof. It shows that the generated archive structure and library metadata are correct, and that the content survives import into two distinct LMS environments. What it does not show is whether the surrounding learning workflow is closed-loop, which is why gradebook passback is still listed as future work in the thesis.

The implementation enforces user authentication with bcrypt password hashing and JWT tokens, validates all request payloads before business logic, parameterises database access through Sequelize, and hardens HTTP responses with Helmet security headers. AI-generated suggestions are treated as untrusted input and rejected if they do not conform to the six supported H5P schemas. This pragmatic baseline addresses the immediate threats inherent in a web tool handling media uploads, user accounts, and AI output, ensuring that malformed content is rejected at the application boundary.

CHAPTER 5

DISCUSSION AND EVALUATION

This chapter discusses the usability study outcomes, compares the platform against the baseline, and evaluates the overall success of the project.

5.1 Discussion

5.2 Comparison

Comparative Evaluation Metrics	
Task completion:	95% vs. 30%.
Median task time:	3.2 min vs. 45 min.
Mean SUS:	82.5 vs. 38.0.
Mean satisfaction:	4.8/5.0 vs. 2.1/5.0.

Figure 5.1: Comparative evaluation: new platform vs. WordPress H5P baseline

Table 5.1 summarises all quantitative metrics from the evaluation.

Taken together, the summary table shows a consistent pattern rather than an isolated win on one metric. The new platform improved completion, time, error recovery, per-

ceived usability, and subjective satisfaction in the same direction, which makes it difficult to explain the results as a sampling artifact or a single favorable task design choice. The metrics reinforce one another and point to the same conclusion about workflow simplification.

Table 5.1: Evaluation metrics summary

Metric	New Platform	WordPress Baseline
Task completion rate	95%	30%
Median task time	3.2 min	45 min
Mean errors per participant	0.9	8.3
Mean SUS score	82.5 / 100	38.0 / 100
Mean satisfaction (Likert)	4.8 / 5.0	2.1 / 5.0

Semi-structured post-session interviews revealed four recurring themes: (1) Seven of ten participants viewed the AI suggestion panel as a primary workflow anchor and adopted an editorial curation model rather than purely manual authoring, consistent with the LLM-as-scaffolding literature. (2) Five participants cited the Bloom’s taxonomy badges as novel pedagogical guidance, leading two participants to explicitly re-balance suggestion acceptance to avoid over-clustering at the “understand” level. This observation confirms the design decision to make taxonomy visible in the suggestion interface. (3) All ten participants preferred the Server-Sent Events streaming interface (where suggestions appeared progressively) over a hypothetical batch interface, aligning with Nielsen’s visibility heuristic. (4) Eight participants spontaneously praised the “Capture timestamp” button, which eliminated the error-prone practice of re-seeking the video position after form transitions. These qualitative observations support the hypothesis that authoring workflow coherence matters more than individual feature polish.

5.3 Evaluation

Evaluation Methodology.

The evaluation employed a within-subject comparative usability study with ten IT students at HCMIU, who completed identical authoring tasks on both the new platform and WordPress H5P plugin in counterbalanced order, with 30-minute rest between sessions. Participants (mean age 21.3 years; 6 male, 4 female; all novices to H5P, 8/10 with minimal WordPress experience, 7/10 with basic web development knowledge) served as educator proxies per standard usability evaluation practice. Task scenarios consisted of: (1) upload a 5-minute video, (2) add ≥ 3 H5P questions at appropriate timestamps, and (3) export the H5P package; on the new platform, participants additionally (4) re-

viewed AI-generated suggestions. Four primary metrics were collected: Task Completion Rate (percentage completing all sub-tasks within 20 minutes), Time-on-Task (seconds from start to valid .h5p package export), Error Count (observable mistakes, correction requirements, or dead-ends), and System Usability Scale (SUS > 68 above average, > 80 excellent). Qualitative data came from think-aloud verbalisations during sessions and semi-structured interviews afterward. The baseline system was WordPress H5P plugin v1.15.8 on standard LAMP stack, with 5-minute orientation equivalent to university ICT induction.

The evaluation results indicate that the main advantage of the platform is not a single algorithmic feature but the removal of workflow friction. Bringing video playback, timestamp capture, question authoring, and AI review into one screen reduced both task time and error recovery effort. The AI feature is useful, but the qualitative results show that it works best as a drafting aid under instructor supervision rather than as an autonomous authoring system.

That interpretation is consistent across the entire chapter. The platform's value comes from composing several small improvements into one coherent workflow: clearer state, fewer screen changes, visible AI assistance, and a direct export path. None of those changes alone explains the observed gains, but together they shift the authoring task from software management back to instructional design.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This thesis has presented the design, development, and evaluation of a cloud-native H5P interactive video content creation platform for university instructors, with a dual focus on user-centred workflow design and AI-assisted question generation.

The core technical contribution is a full-stack web application comprising a React 18 + TypeScript single-page application [37, 38], an Express.js REST API [45], a SQLite database managed via the Sequelize ORM [46, 47], and the Lumieducation H5P server library [15]. Layered on top of this foundation is an AI pipeline that extracts timestamped transcripts from videos (via YouTube captions, uploaded WebVTT files, or Whisper audio transcription [34]), submits them to a large language model (Anthropic Claude [20] or a locally hosted Ollama instance), and returns a Zod-validated list of H5P interaction suggestions classified by Bloom’s revised taxonomy [10]. A Server-Sent Events streaming interface delivers token-level AI progress updates to the browser, providing immediate feedback during the 8–12 second generation latency—consistent with Nielsen’s requirement for system status visibility [22].

The platform was evaluated against the dominant WordPress-based H5P workflow in a comparative usability study with ten participants. The new platform achieved a 95% task completion rate (versus 30% for the baseline), a median task time of 3.2 minutes (versus 45 minutes), a mean error count of 0.9 (versus 8.3), a mean SUS score of 82.5 (versus 38.0), and a mean satisfaction rating of 4.8/5 (versus 2.1/5). These results represent substantial improvements across every metric and exceed all non-functional requirements set at the outset of the project.

RQ1: What are the specific usability barriers in current H5P authoring workflows?

The primary barriers identified were multi-screen navigation fragmentation, the requirement to understand CMS concepts irrelevant to pedagogy, the absence of real-time preview, and the absence of any mechanism for obtaining question suggestions. All four were directly addressed in the new platform.

RQ2: How can user-centred design reduce extraneous cognitive load?

Co-locating the video player, timestamp capture, question editor, and AI suggestion panel on a single screen eliminated the split-attention effect identified by Chandler and Sweller [12]: the extraneous load imposed by navigating between screens is removed when all task-relevant information is co-present. The integrated format reduced the number of required interface interactions from 47 (WordPress) to approximately 8 for a three-question interactive video—a reduction consistent with the 30–50% task-time gains predicted by Paas, Renkl, and Sweller for integrated-format instructional designs [13].

RQ3: What architecture best supports AI-assisted H5P authoring?

The dual-provider AI architecture (Ollama as first choice, Claude as fallback) combined with Zod-based structured output validation—following the output automater pattern described by White et al. [32]—provides a reliable, cost-flexible pipeline. SSE streaming is essential for managing perceived latency during the 8–12 second generation time. The Lumieducation H5P server library [15] provides a robust, specification-compliant H5P layer without CMS dependency, built on the three-tier architecture recommended for institutional web applications [35].

RQ4: To what extent does the platform outperform the baseline?

The platform outperforms the WordPress baseline on every measured dimension: task completion rate by 65 percentage points, time-on-task by 93%, error rate by 89%, SUS score by 44.5 points (from “Poor” to “Excellent” on Brooke’s scale [30]), and satisfaction by 2.7 points on a five-point scale. Given that the Technology Acceptance Model identifies perceived ease of use as a necessary condition for adoption [18], improvements of this magnitude are of direct practical significance for expanding faculty engagement with interactive video pedagogy at VNU-HCM.

This thesis makes the following contributions:

1. **A production-deployable H5P authoring platform.** The platform provides a complete, open-source, CMS-free H5P authoring environment suitable for institutional deployment. It is the first publicly documented system to combine the Lumieducation H5P server library with a modern React SPA and a JWT-secured REST API.
2. **An AI pipeline for H5P question generation.** The transcript-to-H5P suggestion pipeline is the first documented integration of a production-grade LLM (Claude/Ollama) with structured H5P content type generation, Bloom’s taxonomy classification, and runtime schema validation. The pipeline design is modular and

can be adapted to other educational content formats.

3. **Prompt engineering patterns for structured H5P output.** The system prompt, output automater pattern, and Zod validation strategy developed in this project constitute a reusable template for generating structured educational content with LLMs. The specific challenge of ensuring that LLM-generated JSON conforms to H5P library schemas is documented and solved in a reproducible way.
4. **Empirical evidence for user-centred design in EdTech.** The comparative evaluation provides quantitative evidence that a purpose-built, user-centred platform can dramatically outperform a general-purpose CMS-based workflow for educational authoring tasks, contributing to the evidence base for user-centred design in educational technology.
5. **An alignment with Vietnam’s digital education strategy.** The platform supports VNU-HCM’s digital transformation goals by providing an open-source, locally deployable tool that does not depend on commercial licensing and is compatible with the existing MOOC infrastructure.

Several limitations constrain the interpretation of the findings.

Evaluation sample size. Ten participants satisfies Nielsen’s recommendation for detecting the majority of usability problems [22] but is insufficient for parametric hypothesis testing with adequate statistical power. The reported effect sizes are large enough to be practically interpretable in isolation, but confidence intervals and non-parametric significance tests (e.g., Mann–Whitney U) would require a minimum of approximately 30 participants per condition to achieve power ≥ 0.80 .

Proxy participants. IT students were used as educator proxies. While standard practice for early-stage evaluation, this introduces a gap in ecological validity. Faculty members may exhibit different error patterns and different responses to the AI suggestion interface, so the reported SUS score should be treated as an optimistic upper bound rather than a faculty-level estimate.

Vietnamese-language validation. The platform now supports Vietnamese prompts and Vietnamese UI strings, but the reported AI-quality ratings were collected on English transcripts only. No human-rating study was run on Vietnamese lecture data, so Vietnamese transcription accuracy and Vietnamese question quality remain unvalidated in this thesis.

Learner-side reporting. The current implementation supports H5P package export and LTI launch/deep-linking, but it does not yet record student completion scores or attempt data back into the LMS gradebook. This limits the available evidence for learning analytics and outcomes tracking.

Short-term evaluation. The study measured a single-session task. Long-term adoption—whether instructors continue to use the platform after initial exposure, whether

content quality improves over time, and whether the AI pipeline remains relevant as video content evolves—was not assessed.

Single-institution context. The requirements elicitation and participant pool were anchored at HCMIU. Generalisability to other Vietnamese universities, or to non-Vietnamese educational contexts, requires validation.

AI suggestion quality variance. Suggestion quality depends on transcript quality, which degrades with noisy audio or heavy technical vocabulary. While Whisper achieves competitive word-error rates on clean English audio [34], its performance on Vietnamese academic speech—particularly domain-specific technical terminology—has not been evaluated in this project. Vietnamese-language transcription and question generation are therefore important future directions requiring dedicated evaluation. The use of a single expert rater also introduces confirmation bias and prevents the AI-quality scores from being considered double-blind.

Concurrency and storage contention. The system was stress-tested at up to 50 concurrent users, but it was not profiled for write-lock contention under bursty institutional authoring loads. SQLite is appropriate for a single-institution deployment, but higher write concurrency would require a dedicated database benchmark before stronger scalability claims could be made.

6.2 Future work

The following extensions are identified as high-priority directions for future research and development.

Faculty evaluation study. A longitudinal usability study with actual university faculty, measuring adoption over a full semester, would provide ecologically valid evidence for the platform’s educational impact.

Vietnamese language support. Vietnamese prompt handling and UI localisation are now implemented. The next step is a dedicated evaluation on Vietnamese lecture transcripts with bilingual raters and, ideally, improved ASR handling for academic terminology so that Vietnamese question quality can be measured rather than assumed.

Learning analytics integration. Connecting the platform’s export and interaction events with the VNU-HCM MOOC Learning Record Store would enable instructors to observe how students engage with their interactive videos—which questions cause the most replays, which timestamps trigger the most errors, and how interaction patterns correlate with downstream assessment performance. The current system stops at H5P export and LTI launch/deep-linking, so gradebook passback and attempt tracking remain future work. This integration would directly address the learning analytics opportunity identified by Siemens and Long [40], who argue that the data exhaust of interactive

learning systems constitutes an underutilised resource for instructional improvement.

Independent AI review. A follow-up study should use a second rater or a blinded instructional designer review to assess relevance and Bloom’s alignment so that the AI-quality results are not dependent on a single expert opinion.

Collaborative authoring. Adding real-time collaborative editing (multiple instructors working on the same video simultaneously) via Socket.io’s room-based messaging would enable team-based content production. The Socket.io dependency is already included in the backend and the architecture supports this extension.

AI fine-tuning on educational content. The current pipeline uses general-purpose LLMs with prompt engineering. Fine-tuning a smaller model (e.g., a Llama variant) on a corpus of validated H5P question–transcript pairs from the VNU-HCM MOOC platform would improve suggestion quality and reduce generation latency, particularly for the Ollama local inference path.

Asynchronous video processing queue. The current video upload pipeline performs FFmpeg thumbnail generation synchronously in the request handler, causing 220–410 ms response latencies for the upload endpoint. Implementing an asynchronous job queue (e.g., using Bull or BullMQ) would reduce upload endpoint latency and improve system responsiveness under load.

Database scalability benchmark. A dedicated stress test should measure write-lock behaviour in SQLite under simultaneous video uploads, transcript saves, and AI-result persistence so that the migration threshold to a server-grade database can be justified empirically.

Full WCAG 2.1 AA compliance audit. The current platform includes basic accessibility features (keyboard navigation, ARIA labels on interactive elements) but has not undergone a formal WCAG 2.1 AA compliance audit. Achieving full accessibility compliance is essential before the platform can be offered as an institutional standard tool.

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APPENDIX

A.1 SYSTEM SETUP AND API REFERENCE

A.1.1 Runtime Environment

Table 1 lists the software versions used during development and testing.

Table 1: Key runtime components (representative versions)

Component	Notes
Node.js 18 LTS	Backend runtime; required by H5P server package.
React 18 + Vite	Frontend runtime and build tool (TypeScript used throughout).
SQLite (sqlite3)	Lightweight relational store used in development; Sequelize ORM.
AI tooling	Optional: OpenAI/Anthropic APIs, Ollama local inference, faster-whisper for transcription.
Utilities	Nodemailer (email), Helmet (security), CORS and rate-limiting middleware.

A.1.2 Starting the Platform

1. **Install backend dependencies:**

```
1 cd backend && npm install
```

2. **Configure environment:** Copy backend/.env.example to backend/.env and set JWT_SECRET and optionally ANTHROPIC_API_KEY.

3. **Start the backend** (runs on port 5001 by default):

```
1 cd backend && npm run dev
```

4. **Install frontend dependencies:**

```
1 cd "Simple UX UI Design" && npm install
```

5. **Start the frontend** (runs on port 3002 by default):

```
1 cd "Simple UX UI Design" && npm run dev
```

6. Open the application at <http://localhost:3002> in a modern web browser.

A.1.3 Key API Endpoints Reference

Table 2: Key API endpoints

Method	Path	Description
POST	/api/auth/register	Create a new user account.
POST	/api/auth/login	Authenticate and receive JWT.
GET	/api/videos	List authenticated user's videos.
POST	/api/videos	Upload a new video file.
POST	/api/videos/youtube	Import a YouTube video by URL.
GET	/api/videos/:id	Get a single video's metadata.
DELETE	/api/videos/:id	Soft-delete a video.
GET	/api/h5p/:videoId/content	Get H5P interactions for a video.
POST	/api/h5p/:videoId/content	Add an H5P interaction manually.
GET	/api/h5p/:videoId/export	Export H5P package (binary download).
GET	/api/ai/status	Check AI provider availability.
POST	/api/ai/analyze	Analyse transcript (non-streaming).
POST	/api/ai/analyze-stream	Analyse transcript (SSE streaming).
POST	/api/ai/inject	Inject accepted suggestions into H5P.
POST	/api/transcript/youtube	Extract captions from YouTube video.
GET	/api/admin/users	List all users (admin only).
PATCH	/api/admin/users/:id	Update user role or active status (admin only).
POST	/api/admin/users	Create a new account (admin only).
GET	/api/admin/audit-log	Retrieve admin audit log (admin only).
GET	/api/admin/settings	Get all system settings (admin only).
PUT	/api/admin/settings/:key	Update a system setting (admin only).
PUT	/api/users/profile	Update authenticated user's profile.
POST	/api/auth/verify-email	Submit 6-digit email verification code.
POST	/api/auth/forgot-password	Request password reset email.
POST	/api/auth/reset-password	Submit new password with reset token.
GET	/api/lti/launch	LTI 1.3 launch endpoint for LMS integration.